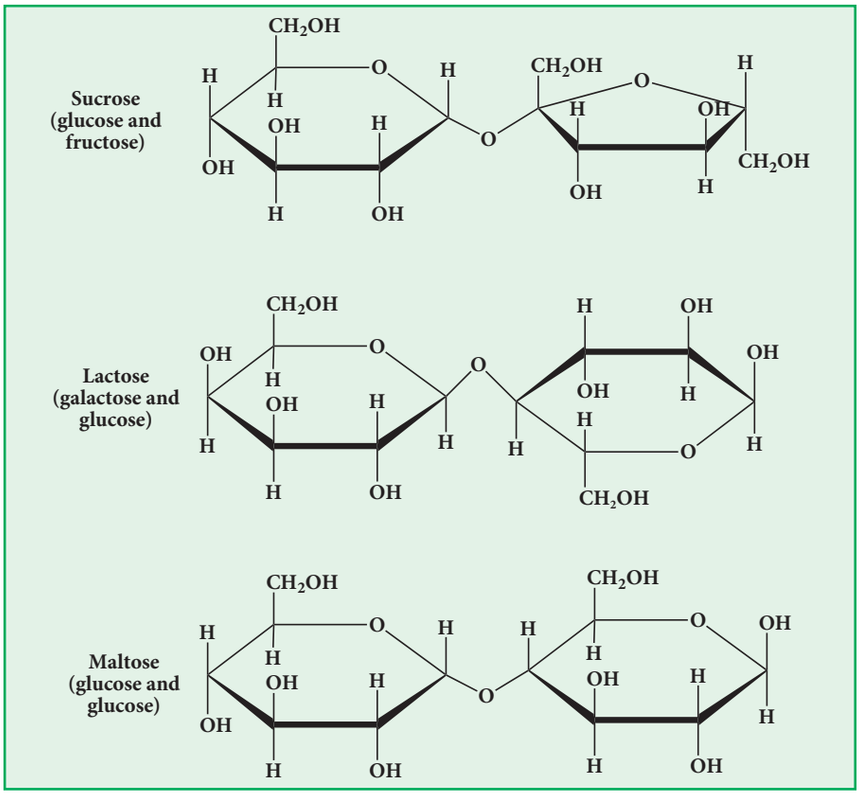


Figure 2.3 Structure of common disaccharides

Hydrolysis of disaccharides in presence of mineral acid yields corresponding monosaccharides. For example hydrolysis of sucrose gives glucose and fructose. Similarly maltose gives two molecules of glucose.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

Sucrose — Glucose + Fructose

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

Maltose — Glucose + Glucose

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

Lactose — Glucose + Galactose

Figure 2.4 Hydrolysis of disaccharides

Trisaccharides

This will give three monosaccharide units on hydrolysis. These have a general formula as $C_n(H_2O)_{n-2}$ Example: raffinose, stachyose.

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